

Calculating the accessibility and spatial availability of jobs (Census Tract level)

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```
library('CommuteCA') # Access data sets from CommuteCA R package
library(dplyr) # A Grammar of Data Manipulation
library(fitdistrplus) # Help to Fit of a Parametric Distribution to Non-Censored or Censored Data
library(scales) # Scale data column-wise in a computationally efficient way
library(here) # enable easy file referencing in project-oriented workflows
library(readr) # Read csv files setting column data types
library(rlang)
library(tidyr) # Create tidy data
library(ggplot2) # Create Elegant Data Visualizations Using the Grammar of Graphics
library(RColorBrewer) # color schemes for maps (and other graphics) designed
library(sf) # support for simple features, a standardized way to encode spatial vector data
library(tmap) # thematic maps
library(nngeo)
```

Introduction

This file is an adaptation of an Rmarkdown from the **CommuteCA** package. CommuteCA was created in conjunction with the office of the *Research Data Center* at *McMaster University*, the *Sherman Centre for Digital Scholarship* and the *Mobilizing Justice*¹.

The **CommuteCA** R package was created to develop standardized methods for transport analysis in research, particularly for analysis using the *2021 Census of Population* from Statistics Canada. We focused our efforts on the *Commuting Reference Guide*, which provides valuable variables and information on commuting for the Canadian population aged 15 and older living in private households.

Considering the number of workers and employment opportunities obtained from the *2021 Census of Population*, using **CommuteCA**, this R markdown aims to create a methodology to obtain Hansen-type accessibility [Hansen, 1959] and spatial accessibility [Soukhov et al., 2023], for all Canadian provinces and territories, considering different modes of travel.

¹ The Mobilizing Justice project is a multidisciplinary and multi-sector collaboration with the objective of understand and address transportation poverty in Canada and to improve the well-being of Canadians at risk of transport poverty. The Social Sciences and Humanities Research Council (SSRHC) has provided funding for the project, which was created by an unprecedented alliance of academics from various Canadian provinces and institutions, transportation firms, and nonprofit organizations

Suggested Readings

- Soukhov, Anastasia, Antonio Paez, Christopher D Higgins, and Moataz Mohamed. 2023."Introducing Spatial Availability, a Singly-Constrained Measure of Competitive Accessibility." *Plos One* 18

(1): e0278468.

- Soukhov, A., Páez, A. (2024). Accessibility analysis for planning applications (Report No. MJ-A2-0002). Mobilizing Justice. <https://github.com/soukhova/MJ-Accessibility-Blogs>
- Pereira, Rafael H. M.; Herszenhut, Daniel. Introduction to urban accessibility: a practical guide with R. Rio de Janeiro: Ipea, 2023. 152 p. ISBN: 978-65-5635-065-3. DOI: <http://dx.doi.org/9786556350653>.

Data

For the Census Tract level analysis, *CommuteCA* provides data sets with land use tables that contains the labour force by transportation mode, and number of jobs in each Census Tract. Information about the file can be seen by:

```
#?land_use_CT_mode
```

We will also use the travel time table and spatial files created in the previous step.

Measuring accessibility

In transportation research, accessibility is a location-based measure² that refers to the potential of a *population* to reach spatially distributed *opportunities*³. These often include destinations that are important to ensure that all members of society can lead plentiful, meaningful lives, such as jobs, parks, cultural activities, health services, education, and, in our case, *jobs*.

For our purposes, the population is the people residing in some origin (a CT) on the date of May 11, 2021⁴, and the opportunities are the jobs located in a given destination (a CT) with which the population interacts. The population uses a mode of transportation (for example, walking, cycling, public transit, car, motorcycle) to reach their destination, with a certain travel cost (distance, time, monetary cost or a combination of several factors).

The result of the accessibility model is a value assigned to each spatial unit - usually at the origin. Areas with high accessibility values have more connections and are closer to more opportunities, while areas with low values have the opposite.

Hansen-type accessibility (HT)

The measure of Hansen [1959] considers that the sum of opportunities in each destination is gradually discounted as the travel cost increases. In this case, opportunities that are easier to access counts

² Location-based metrics measure accessibility as a characteristic of a given location, assuming that all people in the same location have equal access to activities distributed throughout the city. Such measures are sensitive to land use and transportation factors related to the spatial distribution of activities and the configuration and performance of the transportation network. However, they do not take into account the individual characteristics of people.

³ We strongly recommend reading the report by Soukhov and Páez (2024), which provides a detailed explanation of the components of accessibility analysis, as well as their possible uses in equity planning

⁴ Date used as reference for the 2021 Population Census

more for the measure. Many accessibility measures derive from this work [Soukhov and Paez, 2024], according to the equation:

$$HT_i = \sum_{j=1}^J O_j \cdot f(c_{ij})$$

Where:

- i and j : set of spatial units in a region.
- O_j : number of opportunities O at destinations j .
- $f(c_{ij})$: travel cost function, also called decay-function or impedance function.
- HT_i : accessibility score at the origin i being represented by the weighted sum of opportunities considering all destinations O_j .

Impedance functions

The impedance function $f(c_{ij})$ reveals important information about the travel behavior of the population, as it represents the relationship between the “population” at an origin and where they usually go, want to go, or can go to reach the “opportunities” at the destinations. Therefore, defining the impedance function is extremely important.

The decay rate of the impedance function needs to be calibrated if one wants the accessibility estimates to be representative of the people’s travel behavior. In our case, we’ll use the Census data in the calibration process.

Soukhov and Páez [2024] reviewed commonly used impedance functions $f(\bullet)$ in accessibility research, explaining their impacts on the summation of opportunities at specific travel costs c_{ij} . In this study, we will explore the following functions to determine the impedance function that best fits our data: exponential, gamma, log-normal, normal, and uniform.

Spatial Availability

Often, access to opportunities is affected by competition among many people seeking the same opportunity, not just by geographic proximity and transportation costs [Pereira and Herszenhut, 2023]. If competition is not taken into account, it is assumed that the opportunity is equally available to all individuals who seek the opportunity and can reach it - which is not a problem if the opportunity of interest is not exclusive, so that use by one person does not prevent use by another.

Hansen [1959], apart from his innovative implementation of an accessibility measure, does not consider competition in the distribution of opportunities. Also, this measure is considered an unconstrained measure, which means that the same opportunity can be allocated to

different origins, making it difficult to interpret the result. Another difficulty in interpreting the result is caused by the way in which the number of job opportunities is discounted by the cost of travel.

With the goal of including competition in the accessibility measures and constraining the opportunity calculation to a known quantity, [Soukhov et al., 2023] created the *spatial availability* measure. This study also provides a more interpretable result because it guarantees that the measures are summed to a predetermined value, e.g. the total number of jobs in a region, so that each value at the origin can be meaningfully related to this total.

$$SA_i = \sum_{j=1}^J O_j \cdot F_{ij}^t$$

Where:

- SA_i is the number of spatially available opportunities from the perspective of i .
- O_j is the number of opportunities O at the destination j .
- F_{ij} is a balancing factor that depends on the population and cost of travel in the system.

Being $f(c_{ij})$ the impedance function and P_i is the population at origin i , the balancing factor F_{ij} consists of two components:

- F_i^p , a population-based balancing factor that allocate opportunities to i in proportion to the size of the population of the different competing centers. With:

$$F_i^p = \frac{P_i}{\sum_i P_i}$$

- and F_{ij}^c , an impedance-based balancing factor that represents the cost of reaching opportunities. With:

$$F_i^c = \frac{f(c_{ij})}{\sum_i f(c_{ij})}$$

We can obtain F_{ij} by:

$$F_{ij}^t = \frac{F_i^p \cdot F_{ij}^{(c_{ij})}}{\sum_i F_i^p \cdot F_{ij}^{(c_{ij})}}$$

Let's code!

The methodology for obtaining the accessibility measures for Census Tract:

1. First, we load the travel time for each transportation mode. The methodology used to obtain the travel time table is in the previous Rmd.
2. Next, we read the land use table of our study area. This table contains the labour force population and the number of job opportunities for each census tract.
3. We select the impedance functions from our study area.
4. We then apply the impedance functions to the travel time table.
5. We join the travel time and labour force tables, creating the accessibility table. This last table has all the data needed to carry out the accessibility analysis.
6. Finally, we perform the Hansen accessibility type and the spatial availability.

Reading the Land-use table

```
data("land_use_CT_mode")
```

Selecting only CTs from our city of Study:

```
code <- c("3518", # Durham
          "3519", # York
          "3520", # Toronto
          "3521", # Peel
          "3524", # Halton
          "3525" # Hamilton
        )
```

```
code_label <- "GTAH"
```

```
# Filtering by census division
```

```
land_use <- land_use_CT_mode %>%
  filter(PCD %in% code) %>% # Only select respondents who live in the selected PCD
  mutate(PwMode = case_when(PwMode == "Car/motor" ~ "Car-motor",
                             TRUE ~ PwMode))
```

Pre-processing our land use table:

```
# Rename columns
```

```
land_use <- land_use %>%
  rename(jobs = jobs_rounded,
         labour_force = labour_force_est_rounded)
```

```

land_use_exp <- expand.grid(CTUID = unique(land_use$CTUID),
                          PwMode = unique(land_use$PwMode))

land_use_exp <- land_use_exp %>%
  left_join(land_use[,c("CTUID", "PwMode", "labour_force")],
            by = c("CTUID" = "CTUID", "PwMode" = "PwMode")) %>%
  left_join(land_use[,c("CTUID", "PwMode", "jobs")],
            by = c("CTUID" = "CTUID", "PwMode" = "PwMode"))

land_use_exp[is.na(land_use_exp$labour_force),]$labour_force <- 0

land_use_exp <- land_use_exp %>%
  group_by(CTUID) %>%
  mutate(jobs = ifelse(is.na(jobs),
                      max(jobs, na.rm = TRUE), # Use max to get a non-NA value
                      jobs)) %>%
  ungroup()

```

Verifying the number of people within labour force and number of jobs:

```

# labour force
sum(land_use_exp$labour_force)

## [1] 3868120

# Job opportunities
land_use_exp %>%
  group_by(CTUID) %>%
  dplyr::summarize(jobs = first(jobs)) %>%
  dplyr::select(jobs) %>%
  sum(na.rm = TRUE)

## [1] 3043730

```

Impedance functions

All of the impedance functions mentioned require the analyst to define parameters. One useful technique is to create a trip length distribution (TLD) using empirically observed origin-destination travel survey data. A TLD reflects observed travel patterns: specifically, the probability of an observed trip of a given travel cost occurring for the population in a region of interest. Based on the TLD, we can select the most appropriate theoretical PDF forms (e.g., uniform, exponential, gamma), adjust the associated parameters, and use the calibrated theoretical PDF to incorporate the assumptions about the population's travel behavior into the accessibility calculation.

Soukhov and Páez [2024] demonstrated the process of calibrating and selecting the best distribution to represent an impedance function based on travel flows from workers who live and work (full-time) within the City of Hamilton. The authors used the data from the R data package {TTS2016R} and the R package {fitdistrplus} to generate parameters that best fit the TLD the parameters of the uniform, exponential, and gamma functions as closely to the TLD.

We already calibrated impedance functions based on travel time distribution for all Canadian cities, Census Metropolitan Areas and Census Agglomerations, and Province or Territories. We selected the best impedance function based on the on the lowest Akaike Information Criterion (AIC)⁵, selects the better function between the distributions: exponential, gamma, log-normal, normal, and uniform.

⁵ The Akaike Information Criterion (AIC) estimates prediction error and model quality for a given data set.

The chunk below display how to obtain the impedance function for my study area:

```
data("pcd_impedance_functions")

specific_function <- pcd_impedance_functions %>%
  filter(PCD %in% code)

specific_function

## # A tibble: 24 x 10
##       PCD Pr      CMA PwMode distribution  est_1  est_2 loglike    AIC    BIC
##   <dbl> <chr>   <dbl> <chr>   <chr>      <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
## 1  3518 Ontario  996 Car/m~ gamma      1.92   0.0652 -8.57e5  1.71e6  1.71e6
## 2  3518 Ontario  996 Walk   exp       0.0864  0      -2.13e4  4.25e4  4.25e4
## 3  3518 Ontario  996 Trans~ gamma      3.33   0.0591 -6.05e4  1.21e5  1.21e5
## 4  3518 Ontario  535 Bike   gamma      1.89   0.0814 -1.91e3  3.83e3  3.83e3
## 5  3519 Ontario  535 Car/m~ gamma      2.16   0.0764 -1.27e6  2.55e6  2.55e6
## 6  3519 Ontario  535 Trans~ norm     54.9   28.4   -1.00e5  2.01e5  2.01e5
## 7  3519 Ontario  535 Walk   exp       0.0876  0      -2.91e4  5.81e4  5.82e4
## 8  3519 Ontario  535 Bike   lnorm      2.75   0.810  -4.03e3  8.06e3  8.07e3
## 9  3520 Ontario  535 Trans~ gamma      3.60   0.0788 -9.40e5  1.88e6  1.88e6
## 10 3520 Ontario  535 Car/m~ gamma      2.33   0.0861 -1.97e6  3.94e6  3.94e6
## # i 14 more rows
```

In some cases, calibrated functions may not be available for all transportation modes. This is due to RDC vetting rules that prevent the release of data or statistics for spatial units with low populations. However, we also provide datasets at other spatial levels, specifically for Census Metropolitan Areas (CMA)/Census Agglomerations (CA), and Provinces/Territories.

If a city lacks an impedance function for a specific mode, we recommend using the CMA impedance function for that mode. If the city is

within a CMA without a calibrated impedance function or is outside a CMA entirely, we suggest using the provincial impedance function, considering the city's level of metropolitan influence.

```
data("cma_impedance_functions")

CMA_RANGES <- list(
  STRONG_METRO = 996,
  MODERATE_METRO = 997,
  WEAK_METRO = 998,
  NO_METRO = 999,
  TERRITORIES = 0
)

add_new_row <- function(source_df, mode, code, specific_function) {
  new_row <- source_df
  new_row$PCD <- code
  new_row <- new_row %>% dplyr::select(names(specific_function))
  rbind(specific_function, new_row)
}

specific_function_final <- data.frame()

for(pcd_codes in unique(specific_function$PCD)){

  specific_function_pcd <- specific_function %>%
    filter(PCD == pcd_codes)

  if (nrow(specific_function_pcd) < 4) {
    for (mode in c('Bike', 'Walk', 'Car/motor', 'Transit')) {
      if (!(mode %in% specific_function_pcd$PwMode)) {
        cma_value <- specific_function_pcd$CMA[1]

        if (cma_value > 0 && cma_value < 996) {
          cma_row <- cma_impedance_functions %>%
            filter(PwMode == mode, CMA == cma_value)

          if (nrow(cma_row) > 0) {
            specific_function_pcd <- add_new_row(cma_row, mode, code, specific_function_pcd)
          }
        }
      }
    }

    if (nrow(specific_function_pcd) < 4 && cma_value %in% unlist(CMA_RANGES)) {
```



```

cma_type <- switch(
  as.character(cma_value),
  "996" = "Strong metropolitan influenced zone",
  "997" = "Moderate metropolitan influenced zone",
  "998" = "Weak metropolitan influenced zone",
  "999" = "No metropolitan influenced zone",
  "0" = "Territories (outside CAs)"
)

pr_row <- pr_impedance_functions %>%
  filter(PwMode == mode, Pr == specific_function_pcd$Pr[1], CMA_type == cma_type)

if (nrow(pr_row) > 0) {
  pr_row$CMA <- cma_value # Ensure CMA value is consistent
  specific_function_pcd <- add_new_row(pr_row, mode, code, specific_function_pcd)
}
}
}
}

specific_function_final <- rbind(specific_function_final, specific_function_pcd)
}

specific_function_final <- specific_function_final %>%
  mutate(PwMode = case_when(PwMode == "Car/motor" ~ "Car-motor",
                             TRUE ~ PwMode))

```

Reading and preparing the travel matrix table

Reading the travel time table created in the previous rmarkdown[^]. [If you are working with another study area or with the original Census data, don't forget to change the address file for the next chunk.]:

```

ttm_folder <- paste0(here(), "/data-raw/output/", code_label, "/CT/travel_times/")
ttm_file <- paste0(ttm_folder, "ttm.csv")

ttm <- read_csv(ttm_file,
  col_types = cols(
    from_id = col_character(),
    to_id = col_character(),
    travel_time = col_double(),
    PwMode_label = col_character(),
    PCD = col_character(),

```

```

CMAUID = col_character(),
PRUID = col_character()))

ttm[1:10,]

## # A tibble: 10 x 10
##   from_id   to_id travel_time PwMode_label   PCSD CSDNAME PCD   CDNAME CMAUID
##   <chr>     <chr>      <dbl> <chr>      <dbl> <chr>   <chr> <chr>   <chr>
## 1 5320001.00 53200~         0 Bike      3.52e6 Oshawa~ 3518  Durha~ 532
## 2 5320001.00 53200~        13 Bike      3.52e6 Oshawa~ 3518  Durha~ 532
## 3 5320001.00 53200~        23 Bike      3.52e6 Oshawa~ 3518  Durha~ 532
## 4 5320001.00 53200~        38 Bike      3.52e6 Oshawa~ 3518  Durha~ 532
## 5 5320001.00 53200~        49 Bike      3.52e6 Oshawa~ 3518  Durha~ 532
## 6 5320001.00 53200~        53 Bike      3.52e6 Oshawa~ 3518  Durha~ 532
## 7 5320001.00 53200~        32 Bike      3.52e6 Oshawa~ 3518  Durha~ 532
## 8 5320001.00 53200~        37 Bike      3.52e6 Oshawa~ 3518  Durha~ 532
## 9 5320001.00 53200~        30 Bike      3.52e6 Oshawa~ 3518  Durha~ 532
## 10 5320001.00 53200~        19 Bike      3.52e6 Oshawa~ 3518  Durha~ 532
## # i 1 more variable: PRUID <chr>

```

Visualizing some statistics about the travel time matrix:

```

ttm %>%
  group_by(PwMode_label) %>%
  dplyr::summarize(mean_tt = mean(travel_time),
                    median_tt = median(travel_time),
                    min_tt = min(travel_time),
                    max_tt = max(travel_time))

## # A tibble: 4 x 5
##   PwMode_label mean_tt median_tt min_tt max_tt
##   <chr>      <dbl>    <dbl>  <dbl>  <dbl>
## 1 Bike      38.1      40      0     60
## 2 Car-motor 39.0      36      0    120
## 3 Transit   95.0      96      0    149
## 4 Walk      39.0      42      0     60

```

The value of impedance function for each origin-destination-mode will be obtained using travel time of the ttm table and the best impedance function based on the empirical length distributions.

```

generate_impedance <- function(mode_function, travel_cost){

  f <- 0

  # lnorm impedance

```

```

if(mode_function$distribution == "lnorm"){
  f <- dlnorm(travel_cost,
              meanlog = mode_function$est_1,
              sdlog = mode_function$est_2)
}

# Unif impedance
else if(mode_function$distribution == "unif") {
  f <- dunif(travel_cost,
             min = 0,
             max = mode_function$est_2)

# Exponential impedance
} else if(mode_function$distribution == "exp") {
  f <- rescale(dexp(travel_cost,
                   rate = mode_function$est_1))

# Gamma impedance
} else if (mode_function$distribution == "gamma"){
  f <- dgamma(travel_cost,
              shape = mode_function$est_1,
              rate = mode_function$est_2)

# Norm impedance
} else if (mode_function$distribution == "norm"){
  f <- dnorm(travel_cost,
             mean = mode_function$est_1,
             sd = mode_function$est_2)
}

return(f)
}

ttm_f <- data.frame()

ttm[ttm$travel_time == 0,]$travel_time <- 1

for(pcd_codes in unique(ttm$PCD)){
  for(mode in unique(ttm$PwMode_label)){

    ttm_f_mode <- ttm %>%
      filter(PwMode_label == mode,
             PCD == pcd_codes)
  }
}

```

```

pcd_mode_function <- specific_function_final %>%
  filter(PwMode == mode,
         PCD == pcd_codes)

ttm_f_mode$f <- generate_impedance(mode_function = pcd_mode_function,
                                   travel_cost = ttm_f_mode$travel_time)

ttm_f_mode$distribution <- pcd_mode_function$distribution
ttm_f_mode$est_1 <- pcd_mode_function$est_1
ttm_f_mode$est_2 <- pcd_mode_function$est_2

ttm_f <- rbind(ttm_f, ttm_f_mode)
}
}

```

Building the accessibility table

The accessibility table is formed by the junction of the land use and travel times tables:

```

accessibility_table <- ttm_f %>%
  left_join(land_use_exp[,c("CTUID", "PwMode", "labour_force")],
            by = c("from_id" = "CTUID", "PwMode_label" = "PwMode")) %>%
  left_join(land_use_exp[,c("CTUID", "PwMode", "jobs")],
            by = c("to_id" = "CTUID", "PwMode_label" = "PwMode"))

accessibility_table[is.na(accessibility_table$labour_force),]$labour_force <- 0

accessibility_table <- accessibility_table %>%
  group_by(to_id) %>%
  mutate(jobs = ifelse(is.na(jobs),
                      max(jobs, na.rm = TRUE), # Use max to get a non-NA value
                      jobs),
         jobs = ifelse(is.infinite(jobs),
                      0, # Sometimes, a given DA may not have any jobs, resulting in all values in th
                      jobs)) %>%
  ungroup()

```

Confirming the total number of jobs:

```

accessibility_table %>%
  group_by(to_id) %>%
  summarise(job = first(jobs)) %>%
  dplyr::select(job) %>%
  sum()

```

```
## [1] 3043730
```

Checking the total population:

```
accessibility_table %>%
  dplyr::distinct(from_id, PwMode_label, .keep_all = TRUE) %>%
  summarise(pop = sum(labour_force)) %>%
  dplyr::select(pop) %>%
  dplyr::pull(pop) %>%
  sum()
```

```
## [1] 3868040
```

```
CTUID <- land_use_exp %>%
  dplyr::distinct(CTUID, PwMode) %>%
  arrange(CTUID, PwMode) %>%
  rename(id = CTUID, PwMode_label = PwMode)
```

```
from_id <- ttm %>%
  dplyr::distinct(from_id, PwMode_label) %>%
  arrange(from_id, PwMode_label) %>%
  rename(id = from_id)
```

```
setdiff(CTUID, from_id)
```

```
## # A tibble: 1 x 2
##   id          PwMode_label
##   <chr>        <chr>
## 1 5350002.00 Car-motor
```

Calculating the Hansen Accessibility

Generating the Hansen Accessibility (HT):

```
accessibility_table <- accessibility_table %>%
  mutate(jobs_f = jobs * f)
```

```
HT_mode <- accessibility_table %>%
  group_by(from_id, PwMode_label) %>%
  dplyr::summarize(HT_im = sum(jobs_f))
```

The *HT_mode* presents the value of job accessibility for each CT by *mode*. The next chunk sum all *Si* values by CT of origin:

```
HT <- HT_mode %>%
  group_by(from_id) %>%
  dplyr::summarize(HT_i = sum(HT_im))
```

Visualizing the sum of the Hansen Accessibility (HT)⁶:

```
sum(HT$HT_i)

## [1] 83701154
```

⁶ Note that the value itself has no specific meaning, as it is just the sum of the “weighted” jobs, but it can be interpreted as a relative score of the potential interaction based on the observed travel patterns of people residing in the specified region.

Calculating Spatial Availability

Now, we’ll calculate the spatial availability. First, we’ll define the function:

#defining the spatial availability function

```
spatial_availability <- function(df, origin, destination, pop, opp, mode, f){

  origin <- rlang::enquo(origin)
  destination <- rlang::enquo(destination)
  pop <- rlang::enquo(pop)
  opp <- rlang::enquo(opp)
  mode <- rlang::enquo(mode)
  f <- rlang::enquo(f)

  # Calculate sum of population in the system
  sum_pop <- df %>%
    dplyr::distinct(!origin, !mode,
                    .keep_all = TRUE) %>%
    dplyr::summarise(pop = sum(!pop)) %>%
    dplyr::pull(pop) %>%
    sum()

  df$sum_pop <- sum_pop

  # Calculate f_p: population factor
  f_p <- dplyr::pull(df, !pop) / sum_pop

  # Calculate sum of impedance
  sum_impedance <- df %>%
    dplyr::group_by(!destination) %>%
    dplyr::summarize(sum_impedance = sum(!f))

  df <- df %>%
    dplyr::left_join(sum_impedance, by = rlang::as_name(destination))

  # Calculate f_c: impedance factor
```

```

f_c <- dplyr::pull(df, !!f) / df$sum_impedance

df$f_c <- f_c
df$f_p <- f_p

# Calculate f_p * f_c
sum_pa <- df %>%
  dplyr::group_by(!!destination) %>%
  dplyr::summarize(sum_pa = sum(f_p * f_c))

df <- df %>%
  dplyr::left_join(sum_pa,
    by = rlang::as_name(destination))

# Calculate f_t: balancing factor
df$f_t <- (f_p * f_c) / dplyr::pull(df, sum_pa)

# Calculate the Spatial Availability
df %>%
  dplyr::mutate(SA_ij = !!opp * f_t,
    SA_nonc_ij = !!opp * f_c)
}

```

Calculating the spatial availability:

```

# Apply the function to accessibility_table
SA_ij_mode <- accessibility_table %>%
  spatial_availability(origin = from_id,
    destination = to_id,
    pop = labour_force,
    mode = PwMode_label,
    opp = jobs,
    f = f)

SA_ij_mode %>%
  group_by(to_id) %>%
  summarise(job = first(jobs)) %>%
  dplyr::select(job) %>%
  sum()

## [1] 3043730

sum(SA_ij_mode$SA_ij)

## [1] 3043730

```

The table above displays the spatial availability for each combination of origin, destination and mode. The next chunk synthesizes the information for each origin and mode:

```
SA_mode <- SA_ij_mode %>%
  group_by(from_id, PwMode_label) %>%
  summarise(SA_im = sum(SA_ij),
            SA_nonc_im = sum(SA_nonc_ij),
            labour_force = first(labour_force))

sum(SA_mode$labour_force)

## [1] 3868040
```

The *SA_im* presents the value of spatial availability for each CT by *mode*. The next chunk sum all *SA* values by CT of origin:

```
SA <- SA_mode %>%
  group_by(from_id) %>%
  dplyr::summarize(SA_i = sum(SA_im),
                  SA_nonc_i = sum(SA_nonc_im),
                  labour_force = sum(labour_force))
```

Visualizing the sum of spatial availability:

```
sum(SA$SA_i, na.rm = TRUE)

## [1] 3043730
```

Note that the sum of spatial availability matches with the total amount of job opportunities in the system. As explained above, this happens because this metric is constrained.

Now, we will also obtain the spatial availability per person, simply dividing the number of jobs accessible in each transportation mode by the number of people actually taking these modes:

```
lf_pp_mode <- SA_mode %>%
  dplyr::select(-c(SA_nonc_im, SA_im)) %>%
  pivot_wider(names_from = PwMode_label, values_from = labour_force, names_sep = "_", names_prefix = "lf_")
  mutate("lf_Active" = lf_Walk + lf_Bike)

SA_im_mode <- SA_mode %>%
  dplyr::select(-c(SA_nonc_im, labour_force)) %>%
  pivot_wider(names_from = PwMode_label, values_from = SA_im, names_sep = "_", names_prefix = "SA_")
  mutate("SA_Active" = SA_Walk + SA_Bike)
```



```

SA_General <- SA %>%
  rename(SA_General = SA_i,
         lf_General = labour_force) %>%
  dplyr::select(-c(SA_nonc_i))

SA_pp_wide <- lf_pp_mode %>%
  left_join(SA_im_mode, by = "from_id") %>%
  left_join(SA_General, by = "from_id") %>%
  mutate(Walk = SA_Walk/lf_Walk,
         Walk = ifelse(is.nan(Walk), 0, Walk),

         Bike = SA_Bike/lf_Bike,
         Bike = ifelse(is.nan(Bike), 0, Bike),

         Transit = SA_Transit/lf_Transit,
         Transit = ifelse(is.nan(Transit), 0, Transit),

         `Car-motor` = `SA_Car-motor`/`lf_Car-motor`,
         `Car-motor` = ifelse(is.nan(`Car-motor`), 0, `Car-motor`),

         Active = SA_Active/lf_Active,
         Active = ifelse(is.nan(Active), 0, Active),

         General = SA_General/lf_General,
         General = ifelse(is.nan(General), 0, General))

SA_pp_long <- SA_pp_wide %>%
  dplyr::select(c(from_id, Walk, Bike, Transit, `Car-motor`, Active, General)) %>%
  pivot_longer(cols = c(Walk, Bike, Transit, `Car-motor`, Active, General), names_to = "PwMode_label",

```

Visualizing accessibility

Reading the census_tracts geometries:

```

directory_spatial_files <- paste0(here(), "/data-raw/output/", code_label, "/spatial-files/") # Update the
census_tracts <- st_read(paste0(directory_spatial_files, "census_tracts.gpkg"))

## Reading layer 'census_tracts' from data source
##   'C:\Bruno\NSTP\data-raw\output\GTAH\spatial-files\census_tracts.gpkg'
##   using driver 'GPKG'
## Simple feature collection with 1489 features and 11 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -80.24849 ymin: 43.05052 xmax: -78.43698 ymax: 44.39486
## Geodetic CRS:   WGS 84

```

```
census_tracts <- ngeo::st_remove_holes(census_tracts)
```

Reading the census_divisions geometries:

```
census_subdivisions_raw <- st_read(paste0(directory_spatial_files, "census_subdivisions.gpkg"))
```

```
## Reading layer `census_subdivisions' from data source
```

```
##   `C:\Bruno\NSTP\data-raw\output\GTAH\spatial-files\census_subdivisions.gpkg'
```

```
##   using driver `GPKG'
```

```
## Simple feature collection with 25 features and 7 fields
```

```
## Geometry type: MULTIPOLYGON
```

```
## Dimension:      XY
```

```
## Bounding box:  xmin: -80.24849 ymin: 43.05054 xmax: -78.43698 ymax: 44.43323
```

```
## Geodetic CRS:  WGS 84
```

```
census_subdivisions <- ngeo::st_remove_holes(census_subdivisions_raw)
```

Merging the census_tracts with the accessibility measures tables:

```
land_use_mode <- land_use_exp %>%
```

```
  left_join(unique(land_use[,c("CTUID", "CTNAME", "CMAUID", "CMANAME", "PCD", "PRUID")]), by = c("CTUID"
```

```
land_use_general <- land_use_exp %>%
```

```
  group_by(CTUID) %>%
```

```
  summarise(labour_force = sum(labour_force),
```

```
            jobs = first(jobs)) %>%
```

```
  left_join(land_use %>%
```

```
    dplyr::select("CTUID", "CTNAME", "CMAUID",
```

```
                  "CMANAME", "PCD", "PRUID"), by = c('CTUID')) %>%
```

```
  distinct()
```

```
names(census_tracts)[1] <- "CTUID"
```

```
mode_metrics_ct <-
```

```
  expand_grid(CTUID = unique(census_tracts$CTUID),
```

```
             PwMode_label = unique(HT_mode$PwMode_label)) %>%
```

```
  left_join(census_tracts, by = "CTUID") %>%
```

```
  left_join(HT_mode, by = c("CTUID" = "from_id", "PwMode_label" = "PwMode_label")) %>%
```

```
  left_join(SA_mode, by = c("CTUID" = "from_id", "PwMode_label" = "PwMode_label")) %>%
```

```
  left_join(land_use_mode[,c("CTUID", "PwMode")], by = c("CTUID" = "CTUID", "PwMode_label" = "PwMode"))
```

```
mode_metrics_ct <- mode_metrics_ct %>%
```

```
  mutate(SA_im_pp = SA_im/labour_force,
```

```
         SA_im_pp = ifelse(is.nan(SA_im_pp), 0, SA_im_pp))
```

```
mode_metrics_ct_sf <- st_as_sf(mode_metrics_ct, crs = st_crs(census_tracts))
```

Preparing the tables to plot the maps:

```
# Hansen
HT_metrics_ct <- census_tracts %>%
  full_join(HT, by = c("CTUID" = "from_id")) %>%
  full_join(land_use_general[,c("CTUID", "labour_force", "jobs")], by = c("CTUID" = "CTUID"))

# Spatial Availability
SA_metrics_ct <- census_tracts %>%
  full_join(SA, by = c("CTUID" = "from_id")) %>%
  full_join(land_use_general[,c("CTUID", "jobs")], by = c("CTUID" = "CTUID"))
```

General accessibility maps

Creating a directory to export the figures:

```
diretorio_export_figures <- paste0(here(), "/data-raw/output/", code_label, "/CT/accessibility-figures/")

if(!dir.exists(diretorio_export_figures)){
  dir.create(diretorio_export_figures, recursive = TRUE)}
```

Plotting the count of people within the labour force and count of job opportunities in each census_tracts:

```
SA_metrics_ct <- SA_metrics_ct %>%
  mutate(lf_km = labour_force/LANDAREA,
         jobs_km = jobs/LANDAREA)

labour_force_plot <- tm_shape(SA_metrics_ct) +
  tm_polygons("labour_force",
             style = "jenks",
             palette = "Reds",
             title = "Legend",
             border.col = NULL) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5) +
  tm_scale_bar(position = c("right", "bottom"))+
  tm_compass(position = c("left", "top"), size=1.0) +
  tm_layout(legend.outside = TRUE,
            legend.outside.position = "left",
            panel.labels = "Labour force")

labour_force_km_plot <- tm_shape(SA_metrics_ct) +
  tm_polygons("lf_km",
             style = "jenks",
             palette = "Reds",
```

```

        title = "Legend",
        border.col = NULL) +
tm_shape(census_subdivisions) +
tm_borders("black", lwd=0.5) +
tm_scale_bar(position = c("right", "bottom"))+
tm_compass(position = c("left", "top"), size=1.0) +
tm_layout(legend.outside = TRUE,
           legend.outside.position = "left",
           panel.labels = "Labour force per km2")

jobs_plot <- tm_shape(SA_metrics_ct) +
  tm_polygons("jobs",
             style = "jenks",
             palette = "Blues",
             title = "Legend",
             border.col = NULL) +
tm_shape(census_subdivisions) +
tm_borders("black", lwd=0.5) +
tm_scale_bar(position = c("right", "bottom"))+
tm_compass(position = c("left", "top"), size=1.0) +
tm_layout(legend.outside = TRUE,
           legend.outside.position = "left",
           panel.labels = "Job opportunities")

jobs_per_km_plot <- tm_shape(SA_metrics_ct) +
  tm_polygons("jobs_km",
             style = "jenks",
             palette = "Blues",
             title = "Legend",
             border.col = NULL) +
tm_shape(census_subdivisions) +
tm_borders("black", lwd=0.5) +
tm_scale_bar(position = c("right", "bottom"))+
tm_compass(position = c("left", "top"), size=1.0) +
tm_layout(legend.outside = TRUE,
           legend.outside.position = "left",
           panel.labels = "Jobs per km2")

lf_jobs_CT <- tmap_arrange(labour_force_plot, labour_force_km_plot,
                          jobs_plot, jobs_per_km_plot, ncol = 2)

tmap_save(lf_jobs_CT,
          paste0(diretorio_export_figures,"lf_jobs_CT.jpg"),
          width = 20,

```

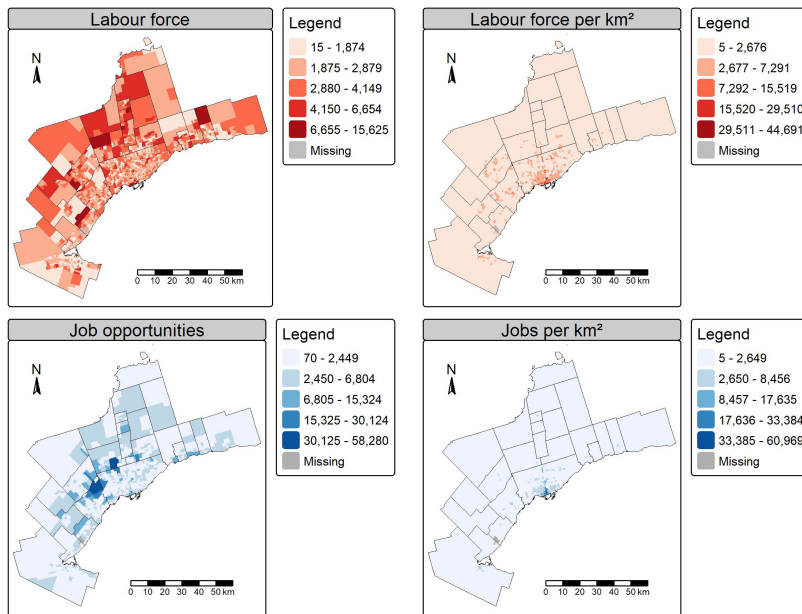
```
height = 15,
units = "cm",
dpi = 300)
```

```
lf_jobs_CT_densities <- tmap_arrange(labour_force_km_plot, jobs_per_km_plot, ncol = 2)
```

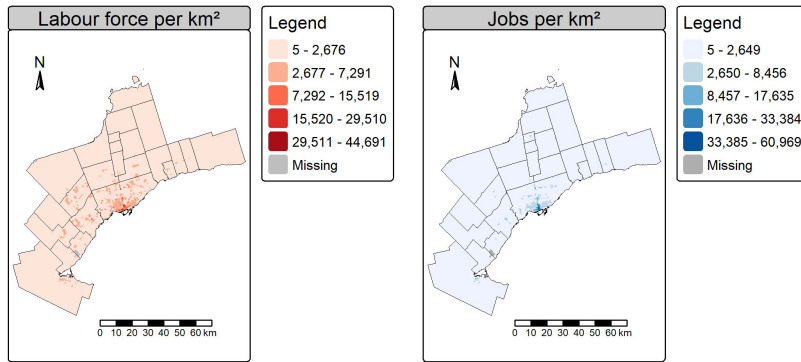
```
tmap_save(lf_jobs_CT,
  paste0(diretorio_export_figures,"lf_jobs_CT.jpg"),
  width = 20,
  height = 15,
  units = "cm",
  dpi = 300)
```

```
tmap_save(lf_jobs_CT_densities,
  paste0(diretorio_export_figures,"lf_jobs_CT_densities.jpg"),
  width = 18,
  height = 8,
  units = "cm",
  dpi = 300)
```

```
knitr::include_graphics(paste0(diretorio_export_figures,"/lf_jobs_CT.jpg"))
```



```
knitr::include_graphics(paste0(diretorio_export_figures,"/lf_jobs_CT_densities.jpg"))
```



Plotting the accessibility measures:

```
ht_plot <- tm_shape(HT_metrics_ct) +
  tm_polygons("HT_i",
    style = "quantile",
    palette = "Purples",
    title = " ",
    border.col = NULL) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5) +
  tm_scale_bar(position = c("right", "bottom"))+
  tm_compass(position = c("left", "top"), size=1.0) +
  tm_layout(legend.outside = TRUE,
    legend.outside.position = c("left","top"),
    panel.labels = "Hansen-type accessibility of jobs")
```

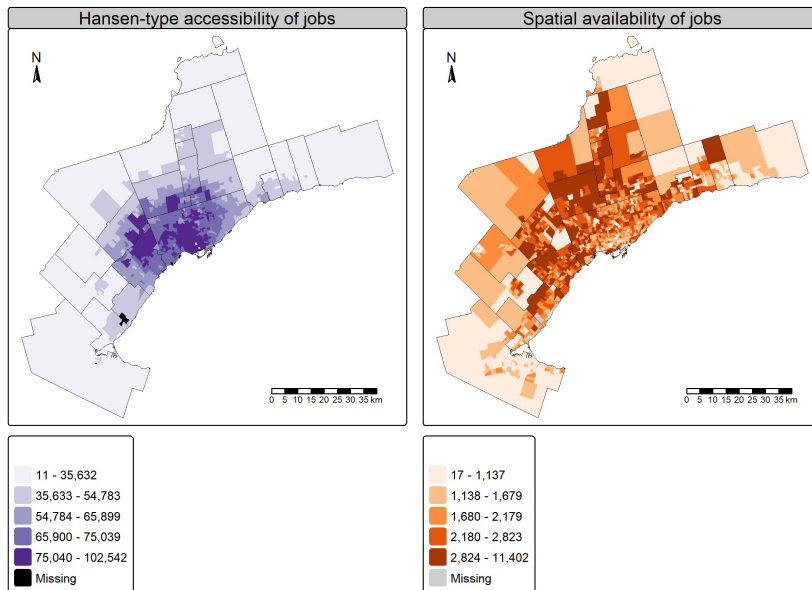
```
sa_plot <- tm_shape(SA_metrics_ct) +
  tm_polygons("SA_i",
    style = "quantile",
    palette = "Oranges",
    title = " ",
    border.col = NULL) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5) +
  tm_scale_bar(position = c("right", "bottom"))+
  tm_compass(position = c("left", "top"), size=1.0) +
  tm_layout(legend.outside = TRUE,
    legend.outside.position = c("left","top"),
    panel.labels = "Spatial availability of jobs")
```

```
HT_SA_CT <- tmap_arrange(ht_plot, sa_plot, ncol = 2)
```

```
tmap_save(HT_SA_CT,
  paste0(diretorio_export_figures,"HT_SA_CT.jpg"),
```

```
width = 20,
height = 15,
units = "cm",
dpi = 300)
```

```
knitr::include_graphics(paste0(diretorio_export_figures, "/HT_SA_CT.jpg"))
```



Plotting the jobs per labour force, and the spatial availability per labour force:

```
HT_metrics_ct$metric <- HT_metrics_ct$jobs / HT_metrics_ct$labour_force
```

```
HT_metrics_ct <- HT_metrics_ct %>%
  mutate(metric = case_when(
    jobs == 0 & labour_force == 0 ~ 0,
    labour_force == 0 & jobs != 0 ~ jobs,
    TRUE ~ metric),
  type = "Jobs per person") %>%
  dplyr::select(CTUID, metric, type)
```

```
SA_metrics_ct$metric <- SA_metrics_ct$SA_i / SA_metrics_ct$labour_force
```

```
SA_metrics_ct <- SA_metrics_ct %>%
  mutate(metric = case_when(
    jobs == 0 & labour_force == 0 ~ 0,
    labour_force == 0 & jobs != 0 ~ jobs,
    TRUE ~ metric),
```

```

  type = "Spatial availability per person") %>%
dplyr::select(CTUID, metric, type)

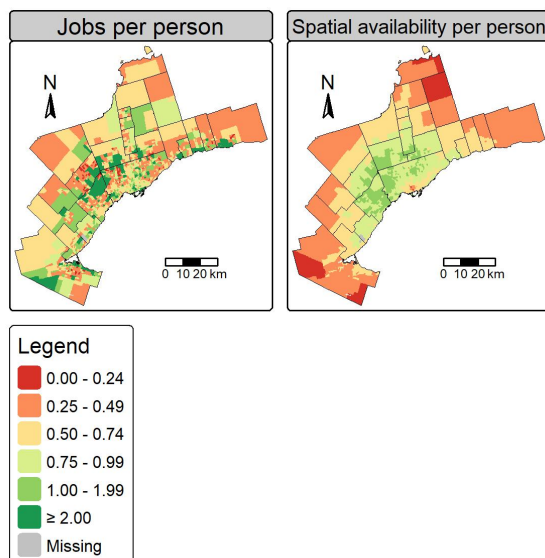
metric_ct <- rbind(HT_metrics_ct, SA_metrics_ct)

SA_per_labour_force <- tm_shape(metric_ct) +
  tm_polygons("metric",
    style = "fixed",
    breaks = c(0, 0.25, 0.5, 0.75, 1, 2, Inf),
    palette = "RdYlGn",
    title = "Legend",
    border.col = NULL) +
  tm_facets(by = "type") +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5) +
  tm_scale_bar(position = c("right", "bottom"), breaks = c(0, 10, 20, 30))+
  tm_compass(position = c("left", "top"), size=1.0)

tmap_save(SA_per_labour_force,
  paste0(diretorio_export_figures,"SA_per_labour_force.jpg"),
  width = 14,
  height = 10,
  units = "cm",
  dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures,"SA_per_labour_force.jpg"))

```



Visualizing the accessibility measures by transportation modes with maps

Visualizing the total labour force by transportation mode:

```
total_pop_by_mode_CT <- tm_shape(mode_metrics_ct_sf) +
  tm_polygons("labour_force",
    style = "cont",
    palette = "Reds",
    title = "Labour force",
    border.col = NULL) +
  tm_facets(by = "PwMode_label", ncol = 2) +
  tm_scale_bar(position = c("right", "bottom"), breaks = c(0, 10, 20, 30))+
  tm_compass(position = c("left", "top"), size = 1.0) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5)

tmap_save(total_pop_by_mode_CT,
  paste0(diretorio_export_figures,"total_pop_by_mode_CT.jpg"),
  width = 14,
  height = 14,
  units = "cm",
  dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures,"total_pop_by_mode_CT.jpg"))
```



```
total_pop_by_mode_CT_2 <- tm_shape(mode_metrics_ct_sf) +
  tm_polygons("labour_force",
    style = "quantile",
    palette = "Reds",
    title = "Labour force",
    border.col = NULL) +
  tm_facets(by = "PwMode_label", ncol = 2) +
  tm_scale_bar(position = c("right", "bottom"), breaks = c(0, 10, 20, 30))+
  tm_compass(position = c("left", "top"), size = 1.0) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5)

tmap_save(total_pop_by_mode_CT_2,
  paste0(diretorio_export_figures,"total_pop_by_mode_CT_2.jpg"),
  width = 14,
  height = 14,
  units = "cm",
  dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures,"total_pop_by_mode_CT_2.jpg"))
```



```

mode_metrics_ct_sf <- mode_metrics_ct_sf %>%
  mutate(lf_pp = labour_force/LANDAREA)

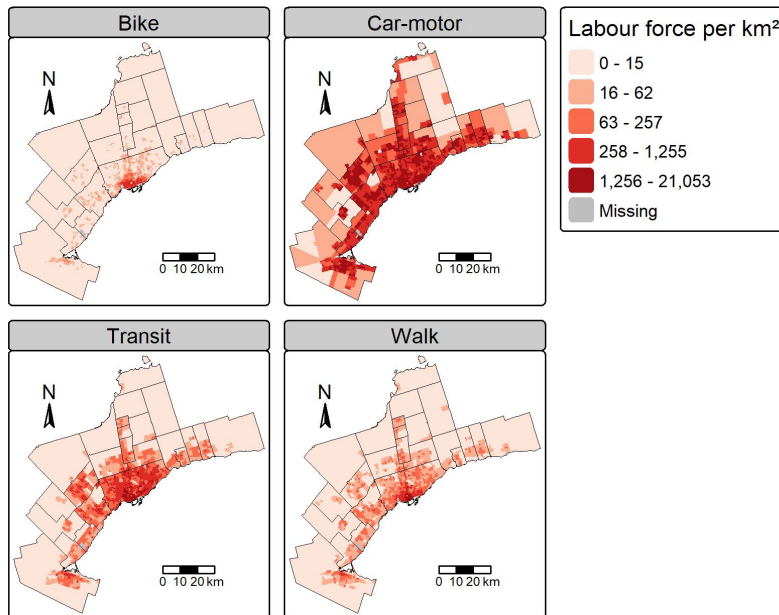
pop_dens_by_mode_CT_2 <- tm_shape(mode_metrics_ct_sf) +
  tm_polygons("lf_pp",
    style = "quantile",
    palette = "Reds",
    title = "Labour force per km2",
    border.col = NULL) +
  tm_facets(by = "PwMode_label", ncol = 2) +
  tm_scale_bar(position = c("right", "bottom"), breaks = c(0, 10, 20, 30))+
  tm_compass(position = c("left", "top"), size = 1.0) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5)

tmap_save(pop_dens_by_mode_CT_2,
  paste0(diretorio_export_figures,"pop_dens_by_mode_CT_2.jpg"),
  width = 14,
  height = 14,
  units = "cm",

```

```
dpi = 300)
```

```
knitr::include_graphics(paste0(diretorio_export_figures,"pop_dens_by_mode_CT_2.jpg"))
```



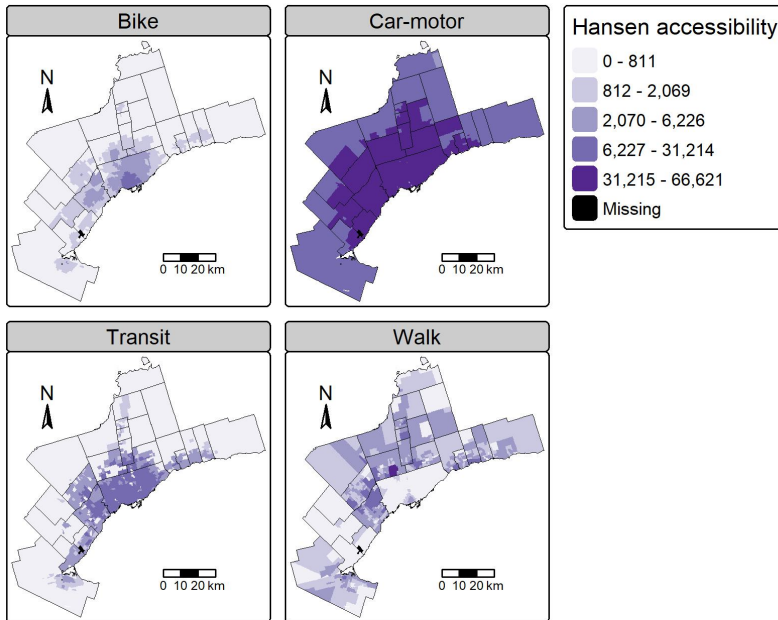
Visualizing the Hansen-type accessibility by transportation mode:

```
HT_by_mode_CT <- tm_shape(mode_metrics_ct_sf) +
  tm_polygons("HT_im",
    style = "quantile",
    palette = "Purples",
    title = "Hansen accessibility",
    border.col = NULL) +
  tm_scale_bar(position = c("right", "bottom"), breaks = c(0, 10, 20, 30))+
  tm_compass(position = c("left", "top"), size=1.0) +
  tm_facets(by = "PwMode_label", ncol = 2) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5)

tmap_save(HT_by_mode_CT,
  paste0(diretorio_export_figures,"HT_by_mode_CT.jpg"),
  width = 14,
  height = 14,
```

```
units = "cm",
dpi = 300)
```

```
knitr::include_graphics(paste0(diretorio_export_figures, "HT_by_mode_CT.jpg"))
```



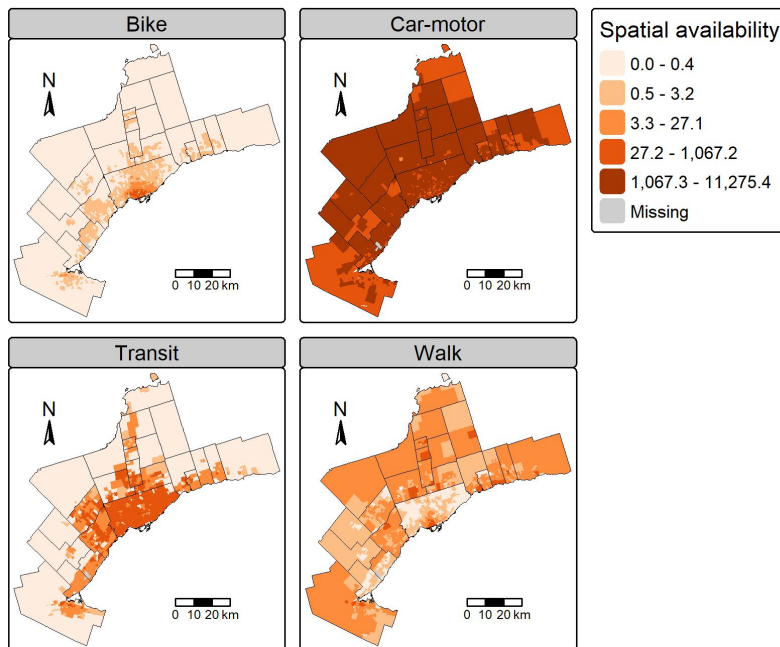
Visualizing the Spatial Availability by transportation mode:

```
SA_by_mode_CT <- tm_shape(mode_metrics_ct_sf) +
  tm_polygons("SA_im",
    style = "quantile",
    palette = "Oranges",
    title = "Spatial availability",
    border.col = NULL) +
  tm_scale_bar(position = c("right", "bottom"), breaks = c(0, 10, 20, 30)) +
  tm_compass(position = c("left", "top"), size=1.0) +
  tm_facets(by = "PwMode_label", ncol = 2) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5)

tmap_save(SA_by_mode_CT,
  paste0(diretorio_export_figures, "SA_by_mode_CT.jpg"),
  width = 14,
```

```
height = 14,
units = "cm",
dpi = 300)
```

```
knitr::include_graphics(paste0(diretorio_export_figures,"SA_by_mode_CT.jpg"))
```

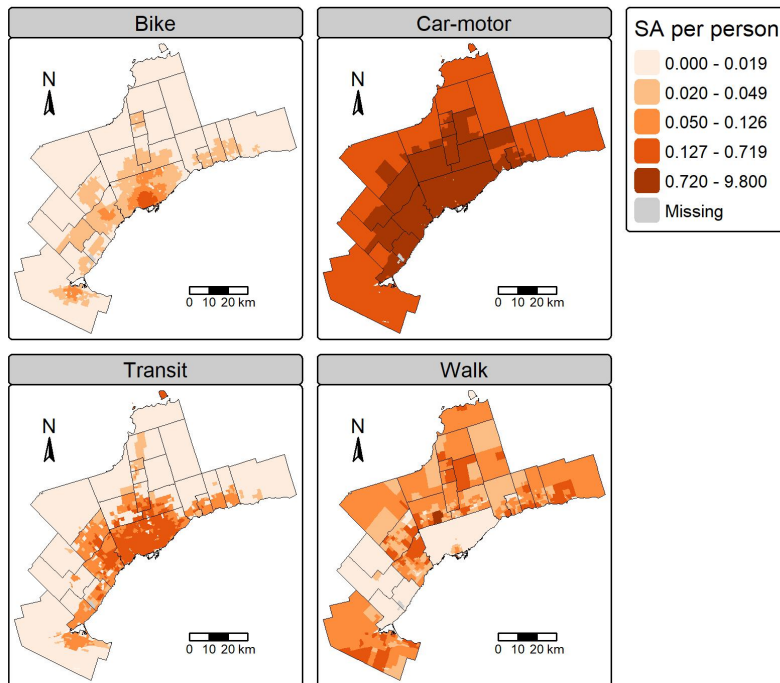


```
SA_per_person_by_mode_CT_2 <- tm_shape(mode_metrics_ct_sf) +
  tm_polygons("SA_im_pp",
    style = "quantile",
    palette = "Oranges",
    title = "SA per person",
    border.col = NULL) +
  tm_scale_bar(position = c("right", "bottom"), breaks = c(0, 10, 20, 30))+
  tm_compass(position = c("left", "top"), size=1.0) +
  tm_facets(by = "PwMode_label", ncol = 2) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5)

tmap_save(SA_per_person_by_mode_CT_2,
  paste0(diretorio_export_figures,"SA_per_person_by_mode.jpg"),
  width = 14,
```

```
height = 14,
units = "cm",
dpi = 300)
```

```
knitr::include_graphics(paste0(diretorio_export_figures,"SA_per_person_by_mode.jpg"))
```



```
SA_pp <- SA_pp_long %>%
  rename(CTUID = from_id)

mode_pp_metrics_ct <-
  expand.grid(CTUID = unique(census_tracts$CTUID),
             PwMode_label = unique(SA_pp$PwMode_label)) %>%
  left_join(census_tracts, by = "CTUID") %>%
  left_join(SA_pp, by = c("CTUID" = "CTUID", "PwMode_label" = "PwMode_label")) %>%
  mutate(SA_pp_im = ifelse(is.nan(SA_pp), 0, SA_pp))

mode_pp_metrics_ct_sf <- st_as_sf(mode_pp_metrics_ct, crs = st_crs(census_tracts))

SA_pp_3modes <- mode_pp_metrics_ct_sf %>%
  filter(PwMode_label %in% c("Active", "Transit", "Car-motor")) %>%
  tm_shape() +
```

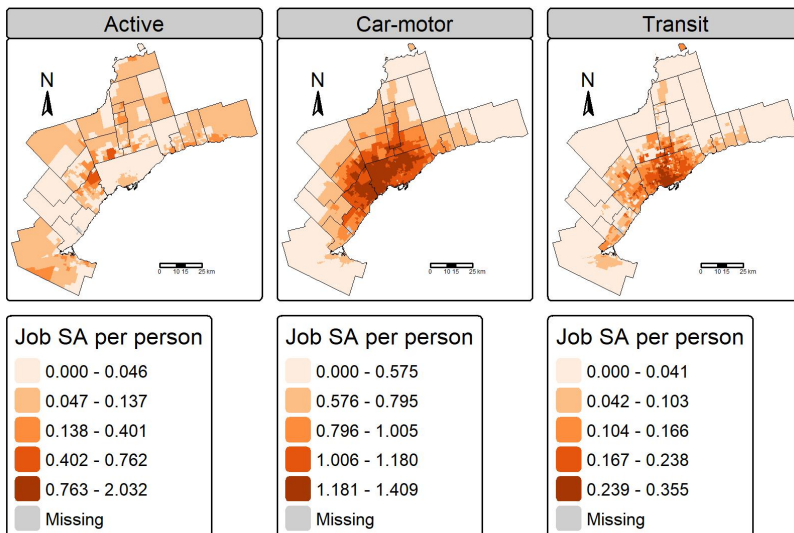
```

tm_polygons("SA_pp_im",
  style = "jenks",
  palette = "Oranges",
  title = "Job SA per person",
  border.col = NULL) +
tm_scale_bar(position = c("right", "bottom"), breaks = c(0, 10, 15, 25)) +
tm_compass(position = c("left", "top"), size=1.0) +
tm_facets(by = "PwMode_label", ncol = 3, free.scales = TRUE) +
tm_shape(census_subdivisions) +
tm_borders("black", lwd=0.5) +
tm_layout(legend.outside = TRUE,
  legend.outside.position = c("right", "bottom"))

tmap_save(SA_pp_3modes,
  paste0(diretorio_export_figures,"SA_pp_3modes.jpg"),
  width = 14,
  height = 10,
  units = "cm",
  dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures,"SA_pp_3modes.jpg"))

```



```

SA_pp_3modes_v2 <- mode_pp_metrics_ct_sf %>%
  filter(PwMode_label %in% c("Active", "Transit", "Car-motor")) %>%
  tm_shape() +
  tm_polygons("SA_pp_im",
    style = "quantile",
    palette = "Oranges",

```



```

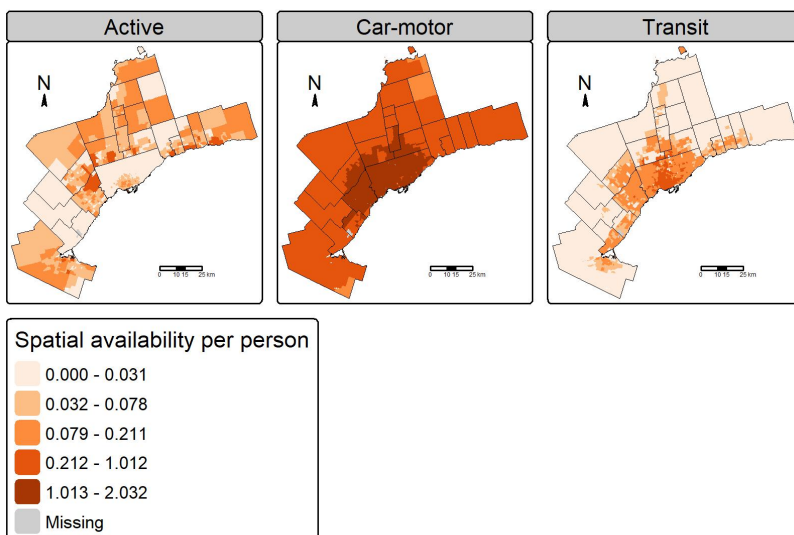
    title = "Spatial availability per person",
    border.col = NULL) +
tm_scale_bar(position = c("right", "bottom"), breaks = c(0, 10, 15, 25)) +
tm_compass(position = c("left", "top"), size=0.5) +
tm_facets(by = "PwMode_label", ncol = 3) +
tm_shape(census_subdivisions) +
tm_borders("black", lwd=0.5) +
tm_layout(legend.outside = TRUE,
           legend.outside.position = c("right", "top"))

tmap_save(SA_pp_3modes_v2,
          paste0(diretorio_export_figures,"SA_pp_3modes_v2.jpg"),
          width = 14,
          height = 10,
          units = "cm",
          dpi = 300)

# For paper
# tmap_save(SA_pp_3modes_v2,
#           paste0(paste0(here::here(),"/manuscript/figures/SA_pp_3modes_v2.jpg")),
#           width = 14,
#           height = 10,
#           units = "cm",
#           dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures,"/SA_pp_3modes_v2.jpg"))

```



Computing of the labour force by transportation mode:

```

total_pop <- sum(land_use_mode$labour_force)
total_SA <- sum(mode_metrics_ct$SA_im, na.rm = TRUE)

modal_SA_percents <- mode_metrics_ct %>%
  group_by(PwMode_label) %>%
  dplyr::summarize(total_labour_fource = sum(labour_force, na.rm = TRUE),
    SA_meam = mean(SA_im, na.rm = TRUE),
    SA_sum = sum(SA_im, na.rm = TRUE),
    perc_SA = sum(SA_im, na.rm = TRUE)/total_SA,
    perc_pop= sum(labour_force, na.rm = TRUE)/total_pop)

modal_SA_long <- modal_SA_percents %>%
  pivot_longer(cols = c(perc_SA, perc_pop), names_to = "perc", values_to = "measure")

```

Visualizing the share of labour force and spatial availability by transportation mode:

```

Pop_SA_by_mode_CT <- ggplot(modal_SA_long,
  aes(x = perc, y = measure, fill = PwMode_label, label = measure %>% percent(accuracy = 0.1))) +
  scale_x_discrete(labels = c(
    "perc_SA" = paste("Spatial Availability\n by Mode\n(Total: ", round(total_SA), ")"),
    "perc_pop" = paste("Population\n by Mode\n(Total: ", format(round(total_pop)), ")") )) +
  geom_col() +
  geom_text(position = position_stack(vjust=0.5), color="black") +
  theme_classic() +
  theme(axis.ticks = element_blank(),
    axis.title = element_blank(),
    axis.line = element_blank(),
    axis.text.y = element_blank(),
    axis.text = element_text(color="black"),
    legend.position = "bottom",
    legend.title = element_blank())

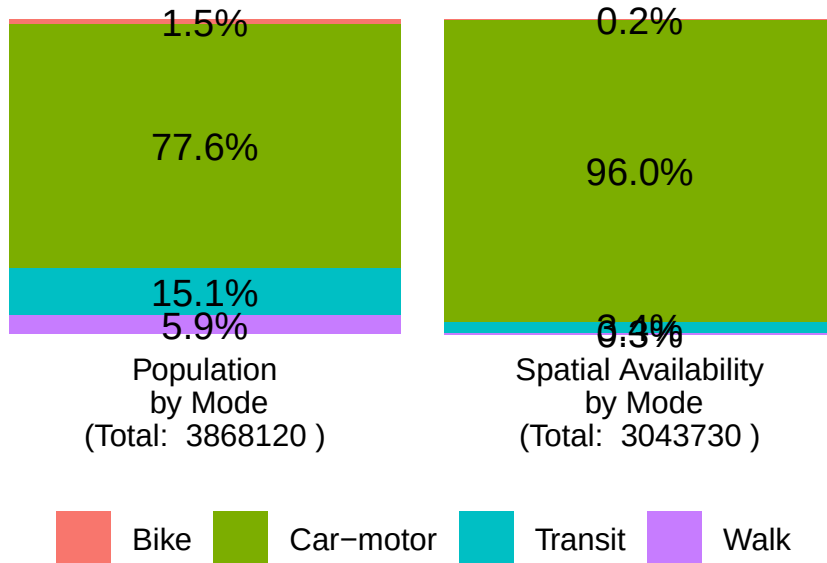
ggsave(file = paste0(diretorio_export_figures,"Pop_SA_by_mode_CT.jpg"),
  plot = Pop_SA_by_mode_CT,
  width = 10,
  height = 10,
  units = "cm",
  dpi = 300)

# # For paper
# tmap_save(Pop_SA_by_mode_CT,
#   paste0(paste0(here::here(), "/manuscript/figures/modal_SA_long.jpg")),
#   width = 10,
#   height = 10,

```

```
#         units = "cm",
#         dpi = 300)
```

Pop_SA_by_mode_CT



Export tables

Creating the directory:

```
diretorio_export <- paste0(here(), "/data-raw/output/", code_label, "/CT/accessibility-measures/") # Update
```

```
if(!dir.exists(diretorio_export)){
  dir.create(diretorio_export, recursive = TRUE)}
```

Export the accessibility files:

```
# HT tables
```

```
write.csv(HT, paste0(diretorio_export, "HT_general.csv"), row.names=FALSE)
write.csv(HT_mode, paste0(diretorio_export, "HT_mode.csv"), row.names=FALSE)
```

```
# SA tables
```

```
write.csv(SA, paste0(diretorio_export, "SA_general.csv"), row.names=FALSE)
write.csv(SA_mode, paste0(diretorio_export, "SA_mode.csv"), row.names=FALSE)
write.csv(SA_pp_wide, paste0(diretorio_export, "SA_pp_wide.csv"), row.names=FALSE)
write.csv(SA_pp_long, paste0(diretorio_export, "SA_pp_long.csv"), row.names=FALSE)
```

```
# Land use table
```

```
write.csv(land_use_mode, paste0(diretorio_export, "land_use_mode.csv"), row.names=FALSE)
```

```
write.csv(land_use_general, paste0(diretorio_export, "land_use_general.csv"), row.names=FALSE)
```

Exporting the `ttm_f` table:

```
write.csv(ttm_f, paste0(ttm_folder, "ttm_f.csv"), row.names=FALSE)
```

Exporting impedance functions used to created the metrics:

```
# write.csv(specific_function_final, paste0(here(), "/manuscript/tables/impedance_functions.csv"), row.names=FALSE)
```

References

- Walter G. Hansen. How accessibility shapes land use. *Journal of the American Institute of Planners*, 25(2):73–76, 05 1959. DOI: 10.1080/01944365908978307. URL <https://doi.org/10.1080/01944365908978307>. `_eprint`: <https://doi.org/10.1080/01944365908978307>.
- Rafael H. M. Pereira and Daniel Herszenhut. *What is accessibility?* Rio de Janeiro, 2023. DOI: 10.38116/9786556350653. URL https://ipeagit.github.io/intro_access_book.
- Anastasia Soukhov and Antonio Paez. Accessibility analysis for planning applications. Technical report, 2024. URL <https://github.com/soukhova/MJ-Accessibility-Blogs>.
- Anastasia Soukhov, Antonio Páez, Christopher D. Higgins, and Moataz Mohamed. Introducing spatial availability, a singly-constrained measure of competitive accessibility. *PLOS ONE*, 18(1):e0278468, 01 2023. DOI: 10.1371/journal.pone.0278468. URL <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0278468>.